# VARUN RAJESH DARJI

**AI Engineer | LLMs, RAG & Agentic Systems | Data Science**

Mumbai, Maharashtra | +91 7977023495 | **varun.darji07@gmail.com** | **LinkedIn** | **Lukey-7**

## SUMMARY

AI Engineer with an M.Sc. in Computer Science (CGPA 8.6) who ships production LLM systems (agentic workflows, RAG pipelines and generative media tools) and backs them with evaluation and testing. At LetsTute, built an AI lead-generation platform drafting outreach at ~$0.04 per lead, a video studio that cut marketing turnaround from ~1 day to ~1 hour, and AI features for a 500+ user ERP. Strong foundation in Python, SQL, statistics and C++, including a quant research engine validated on 11.7 years of market data.

## SKILLS

**AI & LLMs:** RAG, hybrid search & reranking, agentic workflows, LLM evaluation, prompt engineering, LangChain / LangGraph, LlamaIndex, Gemini / Claude / OpenAI APIs

**ML & Data:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, model evaluation, statistical testing, Monte Carlo simulation

**Databases:** SQL (PostgreSQL, SQLite, Supabase), NoSQL (Redis, Neo4j), vector stores (ChromaDB, pgvector)

**Languages & Frameworks:** Python, SQL, C++20, Java, TypeScript, Kotlin; FastAPI, Spring Boot, React, Next.js

**Cloud & DevOps:** Google Cloud Platform (GCP), Docker, GitHub Actions CI/CD, Git, Vercel, Cloudflare R2

## EXPERIENCE

**AI Engineer | LetsTute, Mumbai** Aug 2026 – Present

- Built an AI image and video generation studio (Gemini image models, Veo 3.1, Gemini Omni) that replaced the marketing team's scattered tools with one workflow, cutting asset turnaround from **~1 day to ~1 hour** at an average of **5 videos/day**; supports image-to-video, clip extension and joins with per-second cost rules
- Built **Webex**, the company's LLM lead-generation platform (Next.js, PostgreSQL + pgvector, LangGraph): a supervisor agent delegates lead research and email drafting to worker agents under token budgets and guardrails, producing personalised sequences at **~$0.04/lead**; **1,100+ automated tests**
- Built the OAuth sign-in and Google Ads integration end to end (account sync, reporting, creative-fatigue analysis), feeding a human-in-the-loop agentic ads workflow where no agent grades its own output and failed reviews trigger bounded revisions
- Built AI features for a manufacturing client's ERP serving **500+ users**: bill OCR with a self-hosted vision model plus LLM field extraction with validation guards, policy-document retrieval, and a cited chat assistant, every AI call audit-logged; **240+ tests**
- Built the ERP's identity layer (OIDC provider with PKCE and Google SSO, a sign-in kit shared by every app) and its iOS app, HR employee master and stores item master modules (FastAPI, PostgreSQL, React, Capacitor)

**Java Backend Developer Intern | Tech IT Research, Remote (Fintech)** Dec 2025 – Apr 2026

- Developed Java and Spring Boot backend services for a fintech platform, reducing API response latency by **~33%**
- Integrated the PayPal AddTracking API to automate shipment tracking for **100+ orders**, replacing manual data entry

**Student Intern | Somaiya Vidyavihar University, Mumbai** Jul 2024 – Apr 2025

- Managed and validated **1,000+ student records** for university events and internship drives
- Automated mass email outreach with Google Apps Script, saving **~20 hours** of manual work, and streamlined admin workflows to cut manual effort by **30%** and data errors by **20%**

## PROJECTS

**StudyForge — AI Living Textbook (RAG)** | *Python, FastAPI, React, Supabase, ChromaDB, Gemini* github.com/Lukey-7/StudyForge

- RAG platform that turns PDFs, slides and lectures into a cited, auto-updating textbook plus 16 schema-validated study tools
- 6-layer hybrid retrieval (dense + BM25, RRF, MMR, LLM rerank): on 33 labelled queries, Recall@1 **0.82 → 1.00**, MRR@10 **0.89 → 1.00**; LLM-as-judge check found **95%** of generated paragraphs supported by their sources; **185 tests** and Docker CI

**MARA — Multi-Agent Research Assistant** | *LlamaIndex, Haystack, Hugging Face, ChromaDB, Redis, FastAPI*

- 5-agent pipeline (planner, researcher, summariser, critic, writer) answering over PDFs and web pages with a citation on every factual sentence; notes are kept only if their quote appears verbatim in the source
- Hybrid retrieval with cross-encoder rerank: on 35 labelled queries over 317 chunks, Recall@5 **0.857 → 0.929** and MRR@10 **0.809 → 0.871** vs. dense-only; **108 tests**

**AxiomQuant — Quant Research & Backtesting Engine** | *C++20, Eigen, OpenMP, pybind11, Python* github.com/Lukey-7/AxiomQuant

- Event-driven backtester with realistic costs on **11.7 years** of data and a **606-asset** point-in-time S&P 500 universe; walk-forward testing, block bootstrap, deflated Sharpe and SPA tests showed in-sample Sharpe overstated out-of-sample by **~2.7×** and exposed **~850 pp** of survivorship bias
- Portfolio optimisation (Ledoit-Wolf, constrained QP, risk parity) verified against cvxpy to 1e-8; **50,000-path** OpenMP Monte Carlo up to **7.1×** faster than NumPy; **79 tests** on 4 platforms in CI

**FinTrack — On-Device Personal Finance App** | *Kotlin, Jetpack Compose, Room + SQLCipher, ML Kit, Gemini Nano* github.com/Lukey-7/Personal_Finance_app

- Bank-agnostic SMS parser that turns bank and UPI alerts into transactions, classifies their flow (expense, refund, transfer, investment), deduplicates double-reported payments and learns from a review queue
- Spending insights (trends, subscription and price-rise detection), on-device OCR bill splitting, statement import with balance checks and natural-language queries; **760+ tests** across 5 releases

## EDUCATION

**M.Sc. Computer Science — Somaiya Vidyavihar University, Mumbai** | *CGPA 8.6 | ML coursework: PyTorch, Hugging Face, scikit-learn* 2026

**PG Diploma in Cyber Security — St. Xavier's Technical Institute** 2025

**BCA — Tilak Maharashtra Vidyapeeth** | *CGPA 8.2* 2024